TRANEXAMIC ACID

Trauma: give within 3 h

Slow IV — hypotension

CHEMICAL NAME trans-4-(aminomethyl)cyclohexane-1-carboxylic acid
CLASS Antifibrinolytic (lysine analogue)
TRADE NAMES Cyklokapron · Transamin · Lysteda · Exacyl

THERAPEUTIC USES / INDICATIONS

- Major trauma with significant bleeding (CRASH-2; within 3 h).
- Postpartum haemorrhage (WOMAN trial — within 3 h of birth).
- Surgical blood loss reduction (orthopaedic, cardiac).
- Menorrhagia; epistaxis; dental extraction in haemophilia; hereditary angioedema.

ADMINISTRATION

- **IV** slowly — max 100 mg/min (1 g over 10 min) (rapid → hypotension, seizures).
- **PO** tablets for menorrhagia (during menstruation only).
- **Topical** soaked gauze or mouthwash.

DOSAGES

- **Trauma:** 1 g IV over 10 min within 3 h, then 1 g over 8 h.
- **PPH:** 1 g IV over 10 min; second 1 g after 30 min if bleeding continues.
- **Menorrhagia:** 1–1.3 g PO three times daily for up to 4–5 days.
- **Surgery:** 10–15 mg/kg IV.
- **Children ≥1 yr:** about 20 mg/kg/day (limited data). Reduce in renal impairment (by serum creatinine).

PHARMACOKINETICS

- **Onset:** IV immediate. **Duration:** 7–8 h.
- Minimal metabolism.
- Excreted unchanged in urine. **t½:** 2–3 h.

NURSING MANAGEMENT

- Record time of injury/delivery; benefit only within 3 h.
- Give IV slowly over 10 min; monitor BP.
- Monitor bleeding, Hb, signs of thrombosis.
- Check renal function for repeated dosing.
- Never give via intrathecal route — label syringes clearly.

PHARMACODYNAMICS

- Blocks **lysine-binding sites on plasminogen** → prevents its binding to fibrin → inhibits fibrinolysis, stabilising clots.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting, diarrhoea.
- Dizziness.
- Visual disturbance (colour vision).

ADVERSE EFFECTS (SERIOUS)

- Thromboembolism (DVT, PE, stroke) — rare.
- Seizures (high doses, rapid injection, renal impairment).
- Hypotension with fast IV injection.
- Allergic dermatitis.

CONTRAINDICATIONS & CAUTIONS

- Active thromboembolic disease (DVT, PE, cerebral thrombosis).
- History of convulsions.
- Severe renal impairment (accumulation).
- Subarachnoid haemorrhage (cerebral oedema/infarction); intrathecal administration (fatal).

KEY INTERACTIONS

- **Combined hormonal contraceptives, tretinoin:** ↑ thrombosis risk.
- **Factor IX complex / anti-inhibitor coagulant:** thrombosis.

PATIENT EDUCATION

- Menorrhagia: take only during your period; report leg pain/swelling, chest pain or visual changes.
- Not a contraceptive; avoid with combined pill if at thrombosis risk.

KEY POINTS

- Antifibrinolytic; blocks plasminogen.
- Fast IV → hypotension; seizures at high dose.
- Trauma/PPH: 1 g IV over 10 min within 3 h.
- Avoid with active thrombosis.

SOURCES CHECKED **[435]** Tranexamic Acid 100 mg/ml Solution for Injection (UK SmPC) **[436]** WOMAN trial, Lancet 2017 **[437]** CRASH-2 trial, Lancet 2010 **[438]** Tranexamic Acid, IV (US label) **[439]** Tranexamic Acid, oral (US label)
Full references p. 432–457